

MATH 415-01 Advanced Linear Algebra
Homework #3
Assigned: February 6, 2026
Due: February 13, 2026

1. Determine whether the set of vectors is orthonormal. If it is only orthogonal, normalize the vectors to produce an orthonormal set.

$$\begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}, \quad \begin{bmatrix} 1/3 \\ 2/3 \\ 0 \end{bmatrix}$$

2. Verify that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set, and then find the orthogonal projection of $\mathbf{x}$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

$$\mathbf{x} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}, \quad \mathbf{u}_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

3. Find the closest vector to $\mathbf{x}$ in $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Also find the minimum distance between $\mathbf{x}$ and W .

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \\ 5 \\ 1 \end{bmatrix}, \quad \mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

4. Prove the following theorem from class.

Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then

- (a) $\|U\mathbf{x}\| = \|\mathbf{x}\|$ Hint: calculate $\|U\mathbf{x}\|^2$ and recall how to take the transpose of a product.
- (b) $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$
- (c) $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

5. Let U be a square matrix with orthonormal columns.

- (a) Prove that U is invertible. If you use any theorems from MATH 315, state them. As an example, the Invertible Matrix Theorem is stated below, should you use that theorem.
- (b) Prove that U is an orthogonal matrix.
- (c) [MATH 515] If U is an orthogonal matrix and V is constructed by interchanging some of the columns of U , then use (a) and (b) to prove that V is an orthogonal matrix.

6. Let U and V be $n \times n$ orthogonal matrices. Prove that UV is an orthogonal matrix. (This involves proving that (a) UV is invertible and that (b) its inverse is $(UV)^T$). Hint: remember the rules for taking the inverse or transpose of products of matrices.

7. Given $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$, let $L = \text{Span}\{\mathbf{v}\}$. Prove that the mapping $\mathbf{x} \rightarrow \text{proj}_L(\mathbf{x})$ is a linear transform. (Recall that a linear transformation $T: V \rightarrow W$ satisfies $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$).

8. In class we proved that the columns of an orthogonal matrix U are orthonormal. We will now prove that the rows are orthonormal. We'll break this into steps.

(a) Let's start with a concrete example. Consider the example from class

$$U = \begin{bmatrix} \frac{3}{\sqrt{11}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{66}} \\ \frac{1}{\sqrt{11}} & \frac{2}{\sqrt{6}} & -\frac{4}{\sqrt{66}} \\ \frac{1}{\sqrt{11}} & \frac{1}{\sqrt{6}} & \frac{7}{\sqrt{66}} \end{bmatrix}.$$

Prove that the rows of U are orthonormal.

- (b) Assume that U is an orthogonal matrix. State the definition of an orthogonal matrix. Then either state the definition of an invertible matrix or use the Invertible Matrix Theorem to prove that $UU^T = I$. The Invertible Matrix Theorem is stated below for your convenience.
- (c) Let the rows of U be $\mathbf{r}_i$ for $i = 1, \dots, n$. Then

$$U = \begin{bmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_n \end{bmatrix}.$$

Use (b) to prove that the rows are orthonormal. Hint: use the same approach that we used in class when we proved that the columns are orthonormal. Pay attention to matrix dimensions and matrix multiplication.

9. [MATH 515] Let A be an $m \times n$ matrix. Prove that (a) every $\mathbf{x} \in \mathbb{R}^n$ can be written in the form $\mathbf{x} = \mathbf{p} + \mathbf{u}$, where $\mathbf{p} \in \text{RS}(A)$ and $\mathbf{u} \in \text{NS}(A)$, and (b) if the equation $A\mathbf{x} = \mathbf{b}$ is consistent, then there is a unique $\mathbf{p} \in \text{RS}(A)$ such that $A\mathbf{p} = \mathbf{b}$.

Theorem (The Invertible Matrix Theorem). Let A be an $n \times n$ matrix. The following are equivalent.

- (1) A is an invertible matrix.
- (2) A is row equivalent to the $n \times n$ identity matrix.
- (3) A has n pivot rows.
- (4) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (5) The columns of A form a linearly independent set.
- (6) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one/injective.
- (7) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.
- (8) The columns of A span $\mathbb{R}^n$.
- (9) The linear transformation $x \mapsto A\mathbf{x}$ is onto/surjective.
- (10) There is an $n \times n$ matrix C such that $CA = I$.
- (11) There is an $n \times n$ matrix D such that $AD = I$.
- (12) A^T is an invertible matrix.
- (13) $\det(A) \neq 0$.
- (14) The columns of A form a basis of $\mathbb{R}^n$.
- (15) $\text{CS}(A) = \mathbb{R}^n$.
- (16) $\text{rk}(A) = n$.
- (17) $\nu(A) = 0$.
- (18) $\text{NS}(A) = \{\mathbf{0}\}$.
- (19) The number 0 is not an eigenvalue of A .